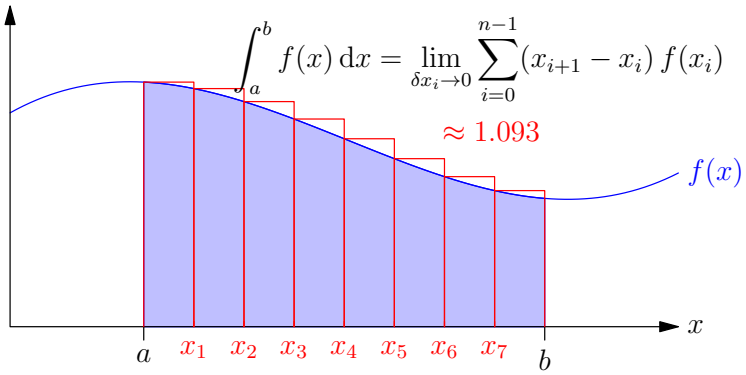
Mathematics of Machine Learning

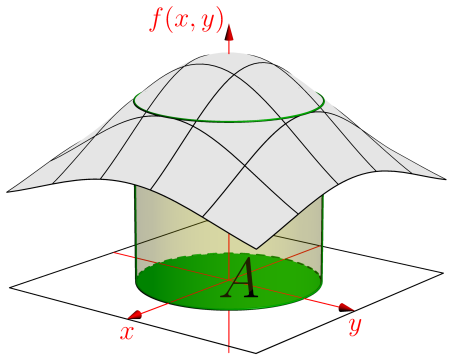
Integral Calculus



Riemann integration, integration by parts, substitution, change of variables

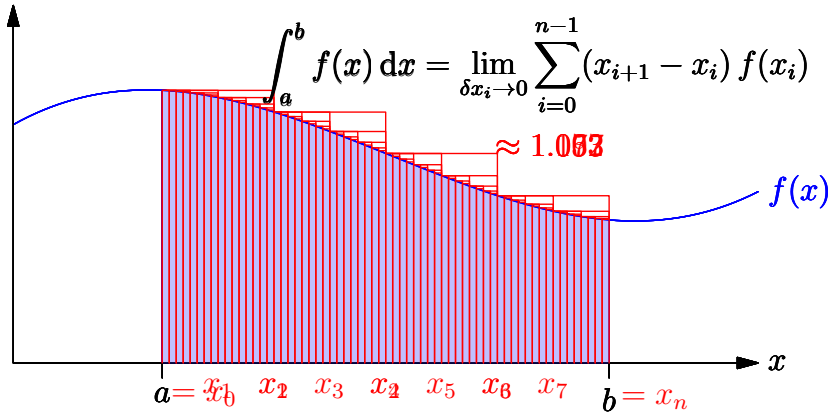
Outline

1. **Defining Integrals**
2. Doing Integrals
3. Changing Variables



Riemann Integral

- Integrals represent area beneath a curve



Linearity of Integration

- Integration is a linear operator

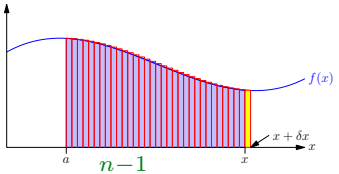
$$\begin{aligned}
\int_a^b (r f(x) + s g(x))\mathrm{d}x &= \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i)(r f(x_i) + s g(x_i)) \\
&= \lim_{\delta x_i \to 0} \left(\sum_{i=0}^{n-1} (x_{i+1} - x_i) r f(x_i) + \sum_{i=0}^{n-1} (x_{i+1} - x_i) s g(x_i) \right) \\
&= \lim_{\delta x_i \to 0} \left(r \sum_{i=0}^{n-1} (x_{i+1} - x_i) f(x_i) + s \sum_{i=0}^{n-1} (x_{i+1} - x_i) g(x_i) \right) \\
&= r \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) f(x_i) + s \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) g(x_i) \\
&= r \int_a^b f(x)\mathrm{d}x + s \int_a^b g(x)\mathrm{d}x
\end{aligned}$$

Fundamental Theorem of Calculus

- Let

$$I(a,x) = \int_a^x f(z)\mathrm{d}z = \lim_{\delta z_i \to 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i)$$

- Now for small δx



$$I(a,x+\delta x) = \int_a^{x+\delta x} f(z)\mathrm{d}z = \lim_{\delta z_i \to 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i) + \delta x f(x)$$

- Thus

$$\frac{\mathrm{d}I(a,x)}{\mathrm{d}x} = \lim_{\delta x \to 0} \frac{I(x+\delta x) - I(x)}{\delta x} = \lim_{\delta x \to 0} \frac{\delta x f(x)}{\delta x} = f(x)$$

The Other Way Around

- Consider

$$\begin{aligned}
\int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}\mathrm{d}x &= \int_a^b \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}\mathrm{d}x \\
&= \lim_{x_{i+1}-x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\
&= \lim_{x_{i+1}-x_i \to 0} \sum_{i=0}^{n-1} (f(x_{i+1}) - f(x_i)) \\
&= (f(x_1) - f(x_0)) + (f(x_2) - f(x_1)) + (f(x_3) - f(x_2)) + \cdots \\
&\quad + (f(x_{n-1}) - f(x_{n-2})) + (f(x_n) - f(x_{n-1})) \\
&= f(x_n) - f(x_0) = f(b) - f(a)
\end{aligned}$$

- We can think of integration as an **anti-derivative**—it undoes differentiation

Indefinite Integrals

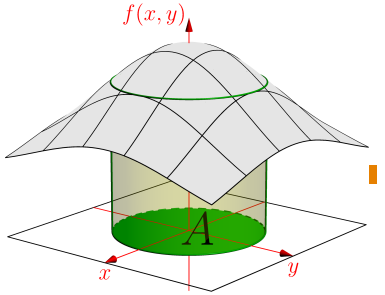
- So far we have considered **definite integrals** where we integrate between two points (a and b)
- However, when think about integration as an anti-derivative, it is useful to think of a function $F(x) = \int f(x)\mathrm{d}x$ (really $\int^x f(z)\mathrm{d}z$)
- So that $F'(x) = f(x)$
- However the function $F(x)$, $F(x)+1$, $F(x)+\pi$, etc. all have the same derivative so $F(x)$ is only defined up to an additive constant
- Note that the definite integral is given by

$$\int_a^b f(x)\mathrm{d}x = F(b) - F(a)$$

Multiple Integrals

- For functions involving many independent variables (e.g. $f(x,y)$, $f(x,y,z)$, $f(\boldsymbol{x})$) we can integrate over multiple dimensions
- For example

$$\iint_A f(x,y)\mathrm{d}x\mathrm{d}y$$

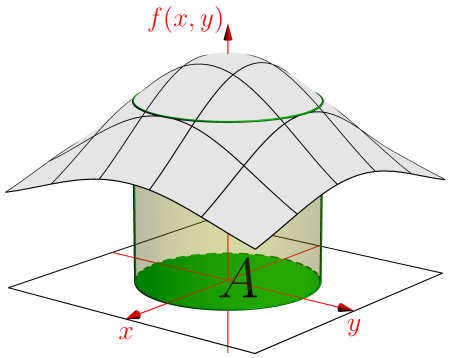


- It gets tedious writing multiple integral signs and I tend to write just one

$$\int \cdots \int f(x_1,x_2,\ldots,x_n)\mathrm{d}x_1\mathrm{d}x_2\cdots\mathrm{d}x_n = \int f(\boldsymbol{x})\mathrm{d}\boldsymbol{x}$$

Outline

1. Defining Integrals
2. **Doing Integrals**
3. Changing Variables



Performing Integration

- A key method for performing integrals is through knowledge of the anti-derivative
- If we know $F'(x) = f(x)$ then $F(x) + c = \int f(x)\mathrm{d}x$
- E.g. we know that $\mathrm{d}x^n/\mathrm{d}x = nx^{n-1}$ therefore

$$\int x^{n-1}\mathrm{d}x = \frac{1}{n}\int \frac{\mathrm{d}x^n}{\mathrm{d}x}\mathrm{d}x = \frac{x^n}{n} + c$$

and

$$\int_a^b x^{n-1}\mathrm{d}x = \frac{b^n}{n} - \frac{a^n}{n}$$

Is Integration Straightforward?

- We saw, due to the product and chain rules, that we can differentiate almost anything. Given integration is the anti-derivative can we integrate anything?
- Products and compositions

$$\int f(x)g(x)\mathrm{d}x = ? \qquad \int f(g(x))\mathrm{d}x = ?$$

- Unfortunately, unlike differentiation we don't have a small parameter we can expand in
- In general integration is hard

Integration by Parts

- Recall the product rule $\frac{\mathrm{d}f(x)g(x)}{\mathrm{d}x} = \frac{\mathrm{d}f(x)}{\mathrm{d}x}g(x) + f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x}$
- Integrating we get

$$\begin{aligned}\int_a^b \frac{\mathrm{d}f(x)g(x)}{\mathrm{d}x}\mathrm{d}x &= \int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}g(x)\mathrm{d}x + \int_a^b f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\mathrm{d}x \\ &= [f(x)g(x)]_a^b = f(b)g(b) - f(a)g(a)\end{aligned}$$

- Unfortunately we get two integrals but we can turn this around

$$\int_a^b f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\mathrm{d}x = [f(x)g(x)]_a^b - \int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}g(x)\mathrm{d}x$$

whether this is helpful depends on $f(x)$ and $g(x)$

Example of Integration by Parts

- Consider

$$\Pi(z) = \int_0^\infty x^z \mathrm{e}^{-x}\mathrm{d}x = \int_0^\infty x^z \frac{\mathrm{d}(-\mathrm{e}^{-x})}{\mathrm{d}x}\mathrm{d}x$$
$$= \left[x^z(-\mathrm{e}^{-x})\right]_0^\infty - \int_0^\infty \frac{\mathrm{d}x^z}{\mathrm{d}x}(-\mathrm{e}^{-x})\mathrm{d}x$$
$$= \int_0^\infty (zx^{z-1})\mathrm{e}^{-x}\mathrm{d}x = z\int_0^\infty x^{z-1}\mathrm{e}^{-x}\mathrm{d}x = z\Pi(z-1)$$

- Thus $\Pi(z) = z\Pi(z-1)$, but

$$\Pi(0) = \int_0^\infty \mathrm{e}^{-x}\mathrm{d}x = \left[-\mathrm{e}^{-x}\right]_0^\infty = -\mathrm{e}^{-\infty} - (-\mathrm{e}^0) = 1$$

- Now
$\Pi(n) = n\Pi(n-1) = n(n-1)\Pi(n-2) = n(n-1)(n-2)\ldots 1 = n!$

Differentiating Through the Integral

- A trick that sometimes works is differentiating through an integral, e.g. consider finding moments

$$M_n = \mathbb{E}[X^n] = \int_{-\infty}^\infty x^n f_X(x)\mathrm{d}x$$

- We can define a moment generating function

$$Z(\ell) = \int_{-\infty}^\infty \mathrm{e}^{\ell x} f_X(x)\mathrm{d}x$$

- Then $M_n = Z^{(n)}(0)$ (the nth derivative of $Z(\ell)$ at $\ell = 0$)

$$\left.\frac{\mathrm{d}^n Z(\ell)}{\mathrm{d}\ell^n}\right|_{\ell=0} = \int_{-\infty}^\infty \left.\frac{\mathrm{d}^n \mathrm{e}^{\ell x}}{\mathrm{d}\ell^n}\right|_{\ell=0} f_X(x)\mathrm{d}x = \int_{-\infty}^\infty x^n f_X(x)\mathrm{d}x = M_n$$

Cumulant Generating Function

- Note that $\mathrm{e}^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \frac{1}{3!}\ell^3 x^3 + \cdots$

- So

$$Z(\ell) = \int_{-\infty}^\infty \mathrm{e}^{\ell x} f_X(x)\mathrm{d}x = 1 + \ell M_1 + \frac{1}{2}\ell^2 M_2 + \frac{1}{3!}\ell^3 M_3 + \cdots$$

- Now using $\log(1+\epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 + \cdots$

$$G(\ell) = \log(Z(\ell)) = \ell M_1 + \frac{1}{2}\ell^2(M_2 - M_1^2) + \frac{1}{3!}\ell^3(M_3 - 3M_2M_1 + 2M_1^3) + \cdots$$

- So that $\kappa_n = G^{(n)}(0)$, with $\kappa_1 = M_1$ (the mean), $\kappa_2 = M_2 - M_1^2$ (the variance), $\kappa_3 = M_3 - 3M_2M_1 + 2M_1^3$ (the third cumulant related to the skewness)

More Integration

- Although we have a few tricks, integration is hard

- Surprisingly integration sometimes is easier when carried out in the complex plane

- This is a beautiful part of mathematics (due largely to Cauchy)—but beyond the scope of this course

- Interestingly, also there is an algorithm that allows us to integrate a lot of function. It is sufficiently complicated that you need to write a computer algorithm of considerable complexity to implement it. Most symbolic manipulation packages (e.g. Mathematica) have implemented some part of this algorithm

Special Functions

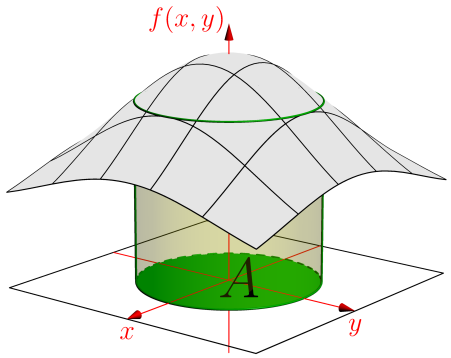
- There are integrals with no known closed form solution
- We saw that $\Pi(z) = \int_0^\infty x^z \mathrm{e}^{-x} \mathrm{d}x$ satisfies $\Pi(z) = z\Pi(z-1)$
- For integer n then $\Pi(n) = n!$, but for general z, the integal $\Pi(z)$ can't be written in terms of elementary functions
- We consider $\Pi(z)$ as a special function in its own right
- Although, history has left us with the gamma function instead

$$\Gamma(z) = \int_0^\infty x^{z-1} \mathrm{e}^{-x} \mathrm{d}x = \Pi(z-1)$$

- Other special function defined by integrals exist (e.g. the Bessel , Aire, hypergeometric, elliptic, error functions, . . .)

Outline

1. Defining Integrals
2. Doing Integrals
3. **Changing Variables**



Substitution

- We can make a transformation from x to $u = u(x)$

$$\begin{aligned}\int_a^b f(x)\mathrm{d}x &= \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} f(x_i)(x_{i+1} - x_i) \\ &= \lim_{\delta u_i \to 0} \sum_{i=0}^{n-1} f(x(u_i)) \frac{x(u_{i+1}) - x(u_i)}{u_{i+1} - u_i}(u_{i+1} - u_i) \\ &= \int_{u(a)}^{u(b)} f(x(u)) \frac{\mathrm{d}x(u)}{\mathrm{d}u} \mathrm{d}u\end{aligned}$$

  - where u_i is such that $x(u_i) = x_i$ or $u_i = u(x_i)$ where $u(x)$ is the inverse of $x(u)$
  - using $\lim_{\delta u_i \to 0} \frac{x(u_{i+1}) - x(u_i)}{u_{i+1} - u_i} = \frac{\mathrm{d}x(u_i)}{\mathrm{d}u}$

Substitution in Practice

- A recipe
  1. Choose $u(x)$: often an "inner" function whose derivative also appears in the integrand
  2. Write $\mathrm{d}u = u'(x)\mathrm{d}x$ and replace every x by u
  3. Change the limits: $x = a$ becomes $u = u(a)$
  4. Integrate in u
- Example: $\int_0^1 2x(1+x^2)^3 \mathrm{d}x$

$$u = 1 + x^2, \quad \mathrm{d}u = 2x\mathrm{d}x, \quad x : 0 \to 1 \;\Rightarrow\; u : 1 \to 2$$

$$\int_0^1 2x(1+x^2)^3 \mathrm{d}x = \int_1^2 u^3 \mathrm{d}u = \left[\frac{u^4}{4}\right]_1^2 = \frac{16-1}{4} = \frac{15}{4}$$

More Substitutions

- The limits can swap over: for $\int_0^{\pi/2} \sin(x)\cos^2(x)\mathrm{d}x$ take $u = \cos(x)$, $\mathrm{d}u = -\sin(x)\mathrm{d}x$, and $x : 0 \to \pi/2$ gives $u : 1 \to 0$

$$\int_0^{\pi/2} \sin(x)\cos^2(x)\mathrm{d}x = \int_1^0 -u^2\mathrm{d}u = \int_0^1 u^2\mathrm{d}u = \frac{1}{3}$$

- A linear change of variable stretches $\mathrm{d}x$: with $z = (x-\mu)/\sigma$, $\mathrm{d}x = \sigma\mathrm{d}z$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}\mathrm{d}x = \sigma\int_{-\infty}^{\infty} \mathrm{e}^{-z^2/2}\mathrm{d}z$$

- The factor σ is the one-dimensional version of the Jacobian determinant (see later slide)

Example of Integration by Substitution

- We consider $I(n) = \int_0^{\infty} x^n \mathrm{e}^{-x^2/2}\mathrm{d}x$

- Let $u(x) = x^2/2$ or $x(u) = \sqrt{2u}$ so that

$$\frac{\mathrm{d}x(u)}{\mathrm{d}u} = \frac{1}{\sqrt{2u}} \qquad u(0) = 0 \qquad u(\infty) = \infty$$

- Thus

$$I(n) = \int_0^{\infty} \left(\sqrt{2u}\right)^n \mathrm{e}^{-u}\frac{1}{\sqrt{2u}}\mathrm{d}u$$
$$= 2^{\frac{n-1}{2}}\int_0^{\infty} u^{\frac{n-1}{2}}\mathrm{e}^{-u}\mathrm{d}u = 2^{\frac{n-1}{2}}\Pi\left(\frac{n-1}{2}\right)$$

- $I(1) = 1$, $I(3) = 2\times 1! = 2$, $I(5) = 2^2\times 2! = 8$, but $I(0) = \Pi(-1/2)/\sqrt{2}$, $I(2) = \sqrt{2}\Pi(1/2) = \Pi(-1/2)/\sqrt{2}$

Gamma Distribution

- Last lecture we met the Gamma distribution

$$\mathrm{Gamma}(x|a,b) = cx^{a-1}\mathrm{e}^{-bx}, \qquad x \geq 0$$

Let's compute the normalisation factor c

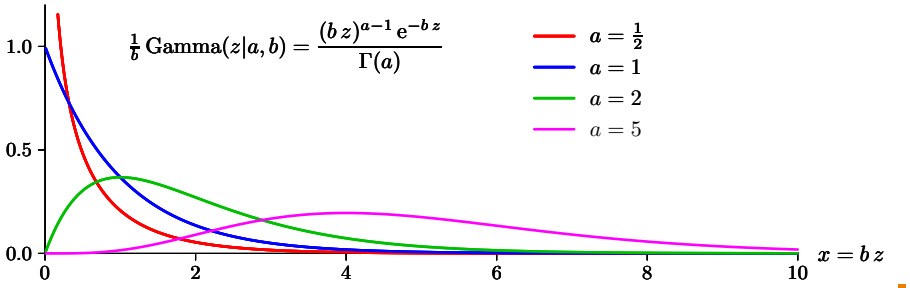
- We need $c\int_0^{\infty} x^{a-1}\mathrm{e}^{-bx}\mathrm{d}x = 1$

- Substitute $z = bx$, so $\mathrm{d}x = \mathrm{d}z/b$ and the limits stay 0 and ∞

$$c\int_0^{\infty}\left(\frac{z}{b}\right)^{a-1}\mathrm{e}^{-z}\frac{\mathrm{d}z}{b} = \frac{c}{b^a}\int_0^{\infty} z^{a-1}\mathrm{e}^{-z}\mathrm{d}z = \frac{c\Gamma(a)}{b^a}$$

- So $c = \dfrac{b^a}{\Gamma(a)} = \dfrac{b^a}{\Pi(a-1)}$: b only sets the scale, a sets the shape

Shapes of the Gamma Distribution

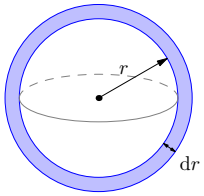


- $a < 1$: infinite at 0; $a = 1$: the exponential distribution; $a > 1$: a peak at $bz = a - 1$

- In terms of $x = bz$ every Gamma distribution is $\mathrm{Gamma}(x|a,1)$: one curve for each a

Lengths of Gaussian Vectors

- Take $\boldsymbol{x} \in \mathbb{R}^n$ with $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$: the density $\propto \mathrm{e}^{-r^2/2}$ depends only on $r = \|\boldsymbol{x}\|$



- The density is largest at $\boldsymbol{x} = \mathbf{0}$, but there is more space at large r: a shell of radius r and thickness $\mathrm{d}r$ has volume $\propto r^{n-1}\mathrm{d}r$
- So, normalising with $I(n-1)$ from the last example,

$$f_R(r) = \frac{r^{n-1}\mathrm{e}^{-r^2/2}}{I(n-1)}, \qquad I(n-1) = 2^{\frac{n}{2}-1}\Gamma\left(\frac{n}{2}\right)$$

- The peak is at $r = \sqrt{n-1}$: in high dimensions almost all the probability is far from the most probable point
- With $s = r^2$, $f_S(s) = \mathrm{Gamma}(s|\frac{n}{2},\frac{1}{2})$: the **chi-squared** distribution with n degrees of freedom

Gamma Distributions Everywhere

- **Chi-squared**: a sum of squares of n independent standard normals is $\mathrm{Gamma}(\frac{n}{2},\frac{1}{2})$
- For m normal samples the sample variance has $(m-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{m-1}$, which lies behind the t-test
- **Maxwell–Boltzmann**: each velocity component of a gas molecule of mass M is $\mathcal{N}(0, kT/M)$, so the speed is the $n = 3$ case, $f(v) \propto v^2\mathrm{e}^{-Mv^2/(2kT)}$, and the energy $\frac{1}{2}Mv^2 \sim \mathrm{Gamma}(\frac{3}{2},\frac{1}{kT})$
- **Waiting times**: $a = 1$ is the exponential distribution; the time to the a-th event of a Poisson process of rate b is $\mathrm{Gamma}(a,b)$
- And it is the conjugate prior for a Poisson rate (Bayes lecture)

Changing Variables in Multidimensional Space

- When changing variables in many dimensions $\boldsymbol{x} \to \boldsymbol{u}$ the change of variables involves the Jacobian

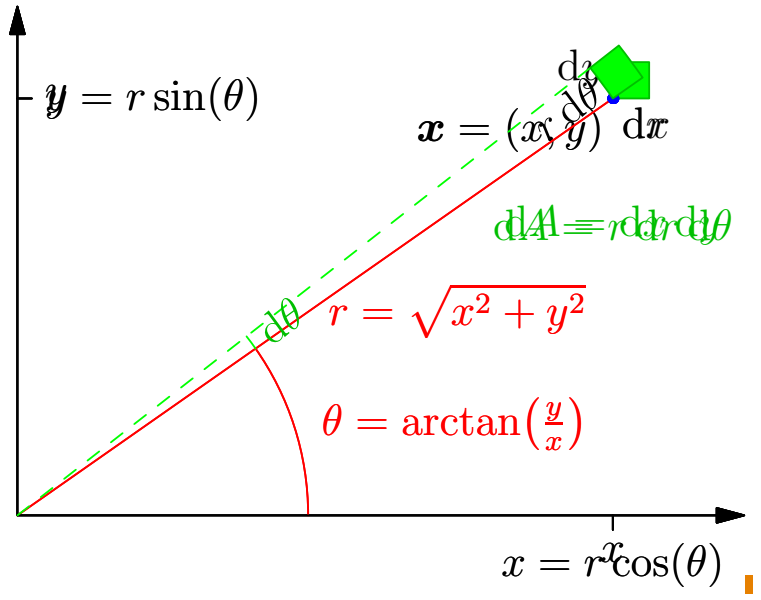
$$\int f(\boldsymbol{x})\mathrm{d}\boldsymbol{x} = \int f(\boldsymbol{x}(\boldsymbol{u}))|\det(\mathbf{J})|\mathrm{d}\boldsymbol{u}, \qquad \boldsymbol{J} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \cdots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_2} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}$$

- E.g. transforming from Cartesian coordinates (x,y) to polar coordinates (r,θ) then $x = r\cos(\theta)$ and $y = r\sin(\theta)$

$$|\det(\mathbf{J})| = \left|\det\begin{pmatrix} \frac{\partial r\cos(\theta)}{\partial r} & \frac{\partial r\cos(\theta)}{\partial \theta} \\ \frac{\partial r\sin(\theta)}{\partial r} & \frac{\partial r\sin(\theta)}{\partial \theta} \end{pmatrix}\right| = \left|\det\begin{pmatrix} \cos(\theta) & -r\sin(\theta) \\ \sin(\theta) & r\cos(\theta) \end{pmatrix}\right|$$
$$= r\left(\cos^2(\theta) + \sin^2(\theta)\right) = r$$

- That is, $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$

Change of Variables in Pictures



Summary

- Integration is extra-ordinarily useful as a tool of analysis
- It occurs when you work with probabilities densities for continuous random variables
- Integration is beautiful, but hard—often impossible
- Changing variables turns many hard integrals into easy ones
- Normal distributions, luckily, give integrals that can be done in closed form, although it takes some work: that is the next lecture
- Making friends with integration will give you a super-power that not too many people share